(10) The modulus function

$$f(x) = |x| = \begin{cases} -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ x & \text{if } x > 0 \end{cases}$$

is continuous at all points of the real line $\mathbb{R}$.

In particular,

$$\lim_{x \to 0^-} |x| = \lim_{x \to 0^-} (-x) = 0,$$

$$\lim_{x \to 0^+} |x| = \lim_{x \to 0^+} (x) = 0, \text{ and}$$

$$\lim_{x \to 0^-} f(x) = 0 = \lim_{x \to 0^+} f(x) = 0 = f(0).$$

(11) The exponential function $f(x) = e^x$ is continuous on $\mathbb{R}$.

(12) The logarithmic function $f(x) = \log x$ $(x > 0)$ in continuous in $(0, \infty)$

9.3.2 Algebra of continuous functions

If f and g are continuous at x_0 then

(1) $f + g$ is continuous at $x = x_0$,

(2) $f - g$ is continuous at $x = x_0$,

(3) $f \,.\, g$ is continuous at $x = x_0$, and

(4) $\dfrac{f}{g}$ is continuous at $x = x_0$ $(g(x) \neq 0)$.

(5) Composite function theorem on continuity.

If f is continuous at $g(x_0)$ and g is continuous at x_0 then fog is continuous at x_0.

Continuity in a closed interval

Definition 9.9

A function $f : [a, b] \to \mathbb{R}$ is said to be continuous on the closed interval $[a, b]$ if it is continuous on the open interval (a, b) and

$$\lim_{x \to a^+} f(x) = f(a) \text{ and } \lim_{x \to b^-} f(x) = f(b).$$

That is, the function f is continuous from the right at a and continuous from the left at b, and is continuous at each point $x_0 \in (a, b)$.

Illustration 9.7

Discuss the continuity of $f(x) = \sqrt{1 - x^2}$.

The domain of definition of f is the closed interval $[-1, 1]$.

(f is defined if $1 - x^2 \geq 0$)

For any point $c \in (-1, 1)$

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